

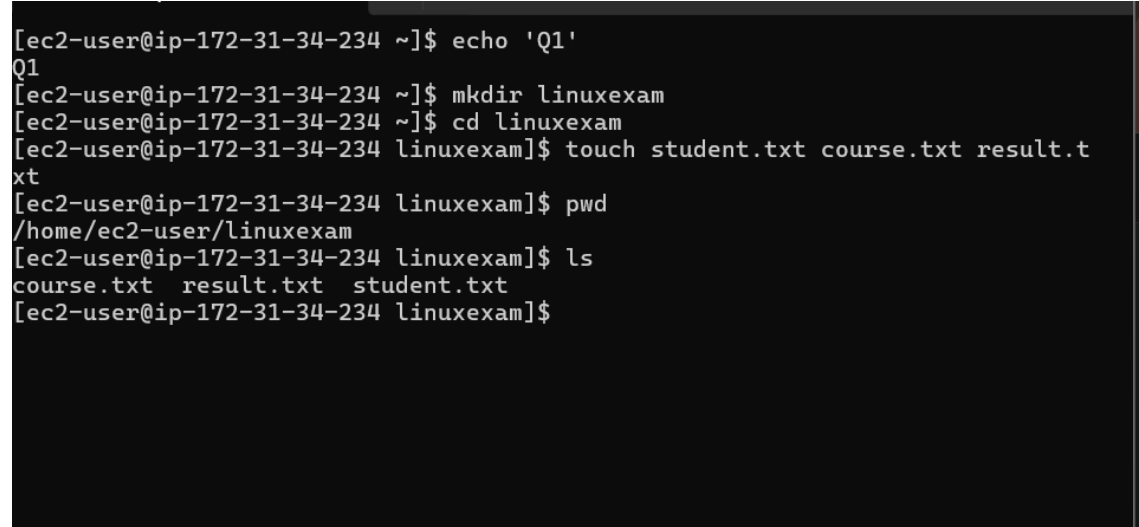
Linux Practical Examination

Q1. Basic File Operations

Commands used:

```
mkdir linuxexam
cd linuxexam
touch student.txt course.txt result.txt
pwd
ls
```

Screenshot:



```
[ec2-user@ip-172-31-34-234 ~]$ echo 'Q1'
Q1
[ec2-user@ip-172-31-34-234 ~]$ mkdir linuxexam
[ec2-user@ip-172-31-34-234 ~]$ cd linuxexam
[ec2-user@ip-172-31-34-234 linuxexam]$ touch student.txt course.txt result.t
xt
[ec2-user@ip-172-31-34-234 linuxexam]$ pwd
/home/ec2-user/linuxexam
[ec2-user@ip-172-31-34-234 linuxexam]$ ls
course.txt  result.txt  student.txt
[ec2-user@ip-172-31-34-234 linuxexam]$
```

Q2. File Management

Commands used:

```
cp student.txt student_backup.txt
mv course.txt linux_course.txt
rm result.txt
mkdir documents backups scripts
mv linuxexam/student_backup.txt backups/
tree
```

Screenshot:

```
[ec2-user@ip-172-31-34-234 linuxexam]$ echo 'Q2'
Q2
[ec2-user@ip-172-31-34-234 linuxexam]$ cp student.txt student_backup.txt
[ec2-user@ip-172-31-34-234 linuxexam]$ mv course.txt linux_course.txt
[ec2-user@ip-172-31-34-234 linuxexam]$ rm result.txt
[ec2-user@ip-172-31-34-234 linuxexam]$ cd
[ec2-user@ip-172-31-34-234 ~]$ mkdir documents backups scripts
[ec2-user@ip-172-31-34-234 ~]$ mv /home/ec2-user/linuxexam /home/ec2-user/backups
[ec2-user@ip-172-31-34-234 ~]$ sudo yum install tree
```

```
4 directories, 3 files
[ec2-user@ip-172-31-34-234 ~]$ mv /home/ec2-user/linuxexam/student_backup.tx
/home/ec2-user/backups
[ec2-user@ip-172-31-34-234 ~]$ tree

├── backups
│   └── student_backup.txt
├── documents
├── linuxexam
│   ├── linux_course.txt
│   └── student.txt
└── scripts

4 directories, 3 files
[ec2-user@ip-172-31-34-234 ~]$
```

Q3. File Content Operations

Commands used:

```
cat > student.txt
cat > linux_course.txt
cat student.txt
head -n 3 student.txt
tail -n 2 student.txt
wc student.txt
```

Screenshot:

```

[ec2-user@ip-172-31-34-234 ~]$ echo 'Q3'
Q3
[ec2-user@ip-172-31-34-234 ~]$ cat > student.txt
hi my name is hanumant
course: mca
collage: srtu
technical skill aws,linux
project: website hosting on aws ec2
^C
[ec2-user@ip-172-31-34-234 ~]$ cat > linux_course.txt
hi my name is vishal
collage: srtu
age: 21
course: mca
project: website hosting
^C
[ec2-user@ip-172-31-34-234 ~]$ cat student.txt
hi my name is hanumant
course: mca
collage: srtu
technical skill aws,linux
project: website hosting on aws ec2
[ec2-user@ip-172-31-34-234 ~]$ cat linux_course
cat: linux_course: No such file or directory
[ec2-user@ip-172-31-34-234 ~]$ cat linux_course.txt
hi my name is vishal
collage: srtu
age: 21
course: mca
project: website hosting
[ec2-user@ip-172-31-34-234 ~]$ head -n 3 student.txt
hi my name is hanumant
course: mca
collage: srtu
[ec2-user@ip-172-31-34-234 ~]$ tail -n 2 student.txt
technical skill aws,linux
project: website hosting on aws ec2
[ec2-user@ip-172-31-34-234 ~]$ wc student.txt
  5  18 112 student.txt
[ec2-user@ip-172-31-34-234 ~]$ |

```

Q4. User Management

Commands used:

```

sudo useradd student01
sudo passwd student01
id student01
eval echo ~student01
su - student01

```

whoami

Screenshot:

```
[ec2-user@ip-172-31-34-234 ~]$ echo 'Q4'
Q4
[ec2-user@ip-172-31-34-234 ~]$ sudo useradd student01
[ec2-user@ip-172-31-34-234 ~]$ sudo passwd student01
Changing password for user student01.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-34-234 ~]$ id student01
uid=1001(student01) gid=1001(student01) groups=1001(student01)
[ec2-user@ip-172-31-34-234 ~]$ getnet passwd student01
-bash: getnet: command not found
[ec2-user@ip-172-31-34-234 ~]$ eval echo ~student01
/home/student01
[ec2-user@ip-172-31-34-234 ~]$ su - student01
Password:
[student01@ip-172-31-34-234 ~]$ whoaim
-bash: whoaim: command not found
[student01@ip-172-31-34-234 ~]$ whoami
student01
[student01@ip-172-31-34-234 ~]$ |
```

Q5. Group Management

Commands used:

```
sudo groupadd linuxbatch
sudo usermod -aG linuxbatch student01
sudo useradd student02
sudo usermod -aG linuxbatch student02
getent group linuxbatch
groups student01
```

Screenshot:

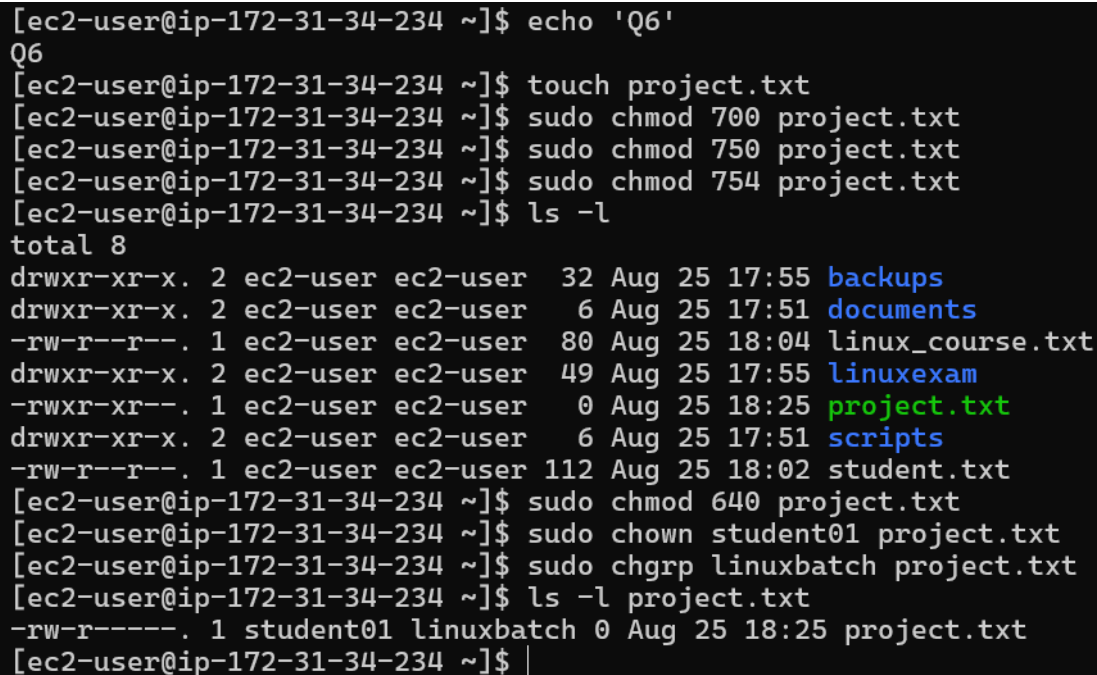
```
[ec2-user@ip-172-31-34-234 ~]$ echo 'Q5'
Q5
[ec2-user@ip-172-31-34-234 ~]$ sudo groupadd linuxbatch
[ec2-user@ip-172-31-34-234 ~]$ sudo usermod -aG linuxbatch student01
[ec2-user@ip-172-31-34-234 ~]$ sudo useradd student02
[ec2-user@ip-172-31-34-234 ~]$ sudo usermod -aG linuxbatch student02
[ec2-user@ip-172-31-34-234 ~]$ getent group linuxbatch
linuxbatch:x:1003:student01,student02
[ec2-user@ip-172-31-34-234 ~]$ groups student01
student01 : student01 linuxbatch
[ec2-user@ip-172-31-34-234 ~]$ |
```

Q6. File Permissions

Commands used:

```
touch project.txt
sudo chmod 754 project.txt
ls -l
sudo chmod 640 project.txt
sudo chown student01 project.txt
sudo chgrp linuxbatch project.txt
ls -l project.txt
```

Screenshot:



```
[ec2-user@ip-172-31-34-234 ~]$ echo 'Q6'
Q6
[ec2-user@ip-172-31-34-234 ~]$ touch project.txt
[ec2-user@ip-172-31-34-234 ~]$ sudo chmod 700 project.txt
[ec2-user@ip-172-31-34-234 ~]$ sudo chmod 750 project.txt
[ec2-user@ip-172-31-34-234 ~]$ sudo chmod 754 project.txt
[ec2-user@ip-172-31-34-234 ~]$ ls -l
total 8
drwxr-xr-x. 2 ec2-user ec2-user 32 Aug 25 17:55 backups
drwxr-xr-x. 2 ec2-user ec2-user  6 Aug 25 17:51 documents
-rw-r--r--. 1 ec2-user ec2-user 80 Aug 25 18:04 linux_course.txt
drwxr-xr-x. 2 ec2-user ec2-user 49 Aug 25 17:55 linuxexam
-rwxr-xr--. 1 ec2-user ec2-user  0 Aug 25 18:25 project.txt
drwxr-xr-x. 2 ec2-user ec2-user  6 Aug 25 17:51 scripts
-rw-r--r--. 1 ec2-user ec2-user 112 Aug 25 18:02 student.txt
[ec2-user@ip-172-31-34-234 ~]$ sudo chmod 640 project.txt
[ec2-user@ip-172-31-34-234 ~]$ sudo chown student01 project.txt
[ec2-user@ip-172-31-34-234 ~]$ sudo chgrp linuxbatch project.txt
[ec2-user@ip-172-31-34-234 ~]$ ls -l project.txt
-rw-r-----. 1 student01 linuxbatch 0 Aug 25 18:25 project.txt
[ec2-user@ip-172-31-34-234 ~]$ |
```

Q7. Permission Challenge

Commands used:

```
mkdir -p linuxexam/public linuxexam/private linuxexam/shared
sudo chmod 777 public
sudo chmod 700 private
sudo chmod 770 shared
ls -ld public private shared
```

Screenshot:

```

[ec2-user@ip-172-31-34-234 ~]$ echo 'Q7'
Q7
[ec2-user@ip-172-31-34-234 ~]$ mkdir -p linuxexam/public linuxexam/private linuxexam/shared
[ec2-user@ip-172-31-34-234 ~]$ ls -l linuxexam
total 0
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 25 17:48 linux_course.txt
drwxr-xr-x. 2 ec2-user ec2-user 6 Aug 25 18:39 private
drwxr-xr-x. 2 ec2-user ec2-user 6 Aug 25 18:39 public
drwxr-xr-x. 2 ec2-user ec2-user 6 Aug 25 18:39 shared
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 25 17:48 student.txt
[ec2-user@ip-172-31-34-234 ~]$ cd linuxexam
[ec2-user@ip-172-31-34-234 linuxexam]$ sudo chmod 777 public
[ec2-user@ip-172-31-34-234 linuxexam]$ sudo chmod 700 private
[ec2-user@ip-172-31-34-234 linuxexam]$ sudo chmod 770 shared
[ec2-user@ip-172-31-34-234 linuxexam]$ ls -ld
drwxr-xr-x. 5 ec2-user ec2-user 92 Aug 25 18:39 .
[ec2-user@ip-172-31-34-234 linuxexam]$ ls -l
total 0
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 25 17:48 linux_course.txt
drwx-----. 2 ec2-user ec2-user 6 Aug 25 18:39 private
drwxrwxrwx. 2 ec2-user ec2-user 6 Aug 25 18:39 public
drwxrwx---. 2 ec2-user ec2-user 6 Aug 25 18:39 shared
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 25 17:48 student.txt
[ec2-user@ip-172-31-34-234 linuxexam]$ ls -ld public private shared
drwx-----. 2 ec2-user ec2-user 6 Aug 25 18:39 private
drwxrwxrwx. 2 ec2-user ec2-user 6 Aug 25 18:39 public
drwxrwx---. 2 ec2-user ec2-user 6 Aug 25 18:39 shared
[ec2-user@ip-172-31-34-234 linuxexam]$ |

```

Q8. Package Management

Commands used:

```

sudo yum update
sudo yum install httpd -y
httpd -v
sudo systemctl start httpd
sudo systemctl status httpd
sudo systemctl enable httpd
sudo systemctl is-enabled httpd

```

Screenshot:

```

[ec2-user@ip-172-31-34-234 linuxexam]$ echo 'Q8'
Q8
[ec2-user@ip-172-31-34-234 linuxexam]$ sudo yum update
Last metadata expiration check: 0:52:41 ago on Tue Aug 25 17:53:00 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-34-234 linuxexam]$ sudo yum install httpd -y
Last metadata expiration check: 0:53:42 ago on Tue Aug 25 17:53:00 2026.
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
httpd x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 46 k
Installing dependencies:
apr x86_64 1.7.5-1.amzn2023.0.4 amazonlinux 129 k
apr-util x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 97 k
apr-util-ldap x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 13 k
generic-logos-httpd noarch 18.0.0-12.amzn2023.0.3 amazonlinux 19 k
httpd-core x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 1.4 M
httpd-filesystem noarch 2.4.68-1.amzn2023.0.1 amazonlinux 12 k
httpd-tools x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 80 k
libbrotli x86_64 1.0.9-4.amzn2023.0.2 amazonlinux 315 k
mailcap noarch 2.1.49-3.amzn2023.0.3 amazonlinux 33 k
Installing weak dependencies:
apr-util-openssl x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 15 k
mod_http2 x86_64 2.0.42-1.amzn2023.0.1 amazonlinux 167 k
mod_lua x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 59 k

Transaction Summary
=====
Install 13 Packages

Total download size: 2.4 M
Installed size: 7.0 M
Downloading Packages:
(1/13): apr-1.7.5-1.amzn2023.0.4.x86_64 2.8 MB/s | 129 kB 00:00
(2/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64 291 kB/s | 13 kB 00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64 1.8 MB/s | 97 kB 00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 582 kB/s | 15 kB 00:00
(5/13): generic-logos-httpd-18.0.0-12.noarch 708 kB/s | 19 kB 00:00
(6/13): httpd-2.4.68-1.amzn2023.0.1.x86_64 1.7 MB/s | 46 kB 00:00
(7/13): httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch 449 kB/s | 12 kB 00:00
(8/13): httpd-tools-2.4.68-1.amzn2023.0.1.x86_64 3.3 MB/s | 80 kB 00:00
(9/13): httpd-core-2.4.68-1.amzn2023.0.1.x86_64 23 MB/s | 1.4 MB 00:00
(10/13): mailcap-2.1.49-3.amzn2023.0.3.noarch 1.0 MB/s | 33 kB 00:00
(11/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7.5 MB/s | 315 kB 00:00
(12/13): mod_lua-2.4.68-1.amzn2023.0.1.x86_64 2.0 MB/s | 59 kB 00:00
(13/13): mod_http2-2.0.42-1.amzn2023.0.1.x86_64 4.7 MB/s | 167 kB 00:00
-----
Total 11 MB/s | 2.4 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Installing : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64 2/13
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
Installing : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
Installing : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64 6/13
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
Running scriptlet: httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch 8/13
Installing : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch 8/13
Installing : httpd-core-2.4.68-1.amzn2023.0.1.x86_64 9/13
Installing : mod_http2-2.0.42-1.amzn2023.0.1.x86_64 10/13
Installing : mod_lua-2.4.68-1.amzn2023.0.1.x86_64 11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.68-1.amzn2023.0.1.x86_64 13/13

```

```

Installing      : httpd-2.4.68-1.amzn2023.0.1.x86_64      13/13
Running scriptlet: httpd-2.4.68-1.amzn2023.0.1.x86_64      13/13
Verifying       : apr-1.7.5-1.amzn2023.0.4.x86_64        1/13
Verifying       : apr-util-1.6.3-1.amzn2023.0.2.x86_64    2/13
Verifying       : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_6 3/13
Verifying       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x8 4/13
Verifying       : generic-logos-httpd-18.0.0-12.amzn2023.0 5/13
Verifying       : httpd-2.4.68-1.amzn2023.0.1.x86_64      6/13
Verifying       : httpd-core-2.4.68-1.amzn2023.0.1.x86_64 7/13
Verifying       : httpd-filesystem-2.4.68-1.amzn2023.0.1.n 8/13
Verifying       : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64 9/13
Verifying       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64   10/13
Verifying       : mailcap-2.1.49-3.amzn2023.0.3.noarch     11/13
Verifying       : mod_http2-2.0.42-1.amzn2023.0.1.x86_64   12/13
Verifying       : mod_lua-2.4.68-1.amzn2023.0.1.x86_64     13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.68-1.amzn2023.0.1.x86_64
httpd-core-2.4.68-1.amzn2023.0.1.x86_64
httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_http2-2.0.42-1.amzn2023.0.1.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-34-234 linuxexam]$ httpd -v
Server version: Apache/2.4.68 (Amazon Linux)
Server built:   Jun  8 2026 00:00:00
[ec2-user@ip-172-31-34-234 linuxexam]$ sudo systemctl start httpd
[ec2-user@ip-172-31-34-234 linuxexam]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled;
   Active: active (running) since Tue 2026-08-25 18:47:32 UTC; 12s
   Docs: man:httpd.service(8)
  Main PID: 28440 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec
   Tasks: 177 (limit: 1059)
  Memory: 13.7M
    CPU: 71ms
  CGroup: /system.slice/httpd.service
          └─28440 /usr/sbin/httpd -DFOREGROUND
            └─28441 /usr/sbin/httpd -DFOREGROUND
              └─28442 /usr/sbin/httpd -DFOREGROUND
                └─28443 /usr/sbin/httpd -DFOREGROUND
                  └─28444 /usr/sbin/httpd -DFOREGROUND

Aug 25 18:47:32 ip-172-31-34-234.ec2.internal systemd[1]: Starting ht
Aug 25 18:47:32 ip-172-31-34-234.ec2.internal systemd[1]: Started ht
Aug 25 18:47:32 ip-172-31-34-234.ec2.internal httpd[28440]: Server co

[ec2-user@ip-172-31-34-234 linuxexam]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-34-234 linuxexam]$ sudo systemctl is-enabled httpd
enabled
[ec2-user@ip-172-31-34-234 linuxexam]$ |

```

Q9. Apache Web Server Configuration

Commands used:

```

sudo nano /etc/httpd/conf/httpd.conf
cd /var/www/html
sudo nano index.html
sudo systemctl restart httpd
curl http://<server-ip>
curl -i http://<server-ip>

```

Screenshot:

```
[ec2-user@ip-172-31-34-234 html]$ echo 'Q9'
Q9
[ec2-user@ip-172-31-34-234 html]$ sudo nano /etc/httpd/conf/httpd.conf
[ec2-user@ip-172-31-34-234 html]$ cd /var/www/html
[ec2-user@ip-172-31-34-234 html]$ sudo nano index.html
[ec2-user@ip-172-31-34-234 html]$ sudo systemctl restart httpd
[ec2-user@ip-172-31-34-234 html]$ curl http://3.95.0.98
student name: hanumant umatawade
roll number: 30
course name: mca
"linux practical examination"

[ec2-user@ip-172-31-34-234 html]$ curl -i http://3.95.0.98
HTTP/1.1 200 OK
Date: Tue, 25 Aug 2026 19:07:05 GMT
Server: Apache/2.4.68 (Amazon Linux)
Last-Modified: Tue, 25 Aug 2026 19:04:30 GMT
ETag: "61-659e3c5c2d370"
Accept-Ranges: bytes
Content-Length: 97
Content-Type: text/html; charset=UTF-8

student name: hanumant umatawade
roll number: 30
course name: mca
"linux practical examination"

[ec2-user@ip-172-31-34-234 html]$ |
```

Q10. Process Management

Commands used:

```
ps -ef
top
pgrep httpd
ps -p <PID> -f
sudo systemctl stop httpd
sudo systemctl status httpd
sudo systemctl start httpd
sudo systemctl status httpd
```

Screenshot:

```
[ec2-user@ip-172-31-34-234 ~]$ echo 'Q10'
```

```
Q10
```

```
[ec2-user@ip-172-31-34-234 ~]$ ps -ef
```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	17:45	?	00:00:01	/usr/lib/syst
root	2	0	0	17:45	?	00:00:00	[kthreadd]
root	3	2	0	17:45	?	00:00:00	[pool_workque
root	4	2	0	17:45	?	00:00:00	[kworker/R-rc
root	5	2	0	17:45	?	00:00:00	[kworker/R-sy
root	6	2	0	17:45	?	00:00:00	[kworker/R-kv
root	7	2	0	17:45	?	00:00:00	[kworker/R-sl
root	8	2	0	17:45	?	00:00:00	[kworker/R-ne
root	10	2	0	17:45	?	00:00:00	[kworker/0:0H
root	12	2	0	17:45	?	00:00:00	[kworker/u8:0
root	13	2	0	17:45	?	00:00:00	[kworker/R-mm
root	14	2	0	17:45	?	00:00:00	[ksoftirqd/0]
root	15	2	0	17:45	?	00:00:00	[rcu_preempt]
root	16	2	0	17:45	?	00:00:00	[rcu_exp_par_
root	17	2	0	17:45	?	00:00:00	[rcu_exp_gp_k
root	18	2	0	17:45	?	00:00:00	[migration/0]
root	19	2	0	17:45	?	00:00:00	[cpuhp/0]
root	20	2	0	17:45	?	00:00:00	[cpuhp/1]
root	21	2	0	17:45	?	00:00:00	[migration/1]
root	22	2	0	17:45	?	00:00:00	[ksoftirqd/1]
root	24	2	0	17:45	?	00:00:00	[kworker/1:0H
root	27	2	0	17:45	?	00:00:00	[kdevtmpfs]
root	28	2	0	17:45	?	00:00:00	[kworker/R-in
root	29	2	0	17:45	?	00:00:00	[rcu_tasks_kt
root	30	2	0	17:45	?	00:00:00	[rcu_tasks_ru
root	31	2	0	17:45	?	00:00:00	[rcu_tasks_tr
root	32	2	0	17:45	?	00:00:00	[kauditd]
root	34	2	0	17:45	?	00:00:00	[khungtaskd]
root	35	2	0	17:45	?	00:00:00	[oom_reaper]
root	37	2	0	17:45	?	00:00:00	[kworker/R-wr
root	38	2	0	17:45	?	00:00:00	[kcompactd0]
root	39	2	0	17:45	?	00:00:00	[khugepaged]
root	40	2	0	17:45	?	00:00:00	[kworker/R-kb
root	41	2	0	17:45	?	00:00:00	[kworker/R-bl
root	42	2	0	17:45	?	00:00:00	[kworker/R-ki
root	43	2	0	17:45	?	00:00:00	[irq/9-acpi]
root	45	2	0	17:45	?	00:00:00	[kworker/R-tp
root	46	2	0	17:45	?	00:00:00	[kworker/R-md
root	47	2	0	17:45	?	00:00:00	[kworker/R-ed
root	48	2	0	17:45	?	00:00:00	[watchdogd]
root	49	2	0	17:45	?	00:00:00	[kworker/R-qu
root	51	2	0	17:45	?	00:00:00	[kworker/1:1H
root	60	2	0	17:45	?	00:00:00	[kswapd0]
root	78	2	0	17:45	?	00:00:00	[kworker/R-xf
root	79	2	0	17:45	?	00:00:00	[kworker/R-xf
root	80	2	0	17:45	?	00:00:00	[kworker/R-cr
root	81	2	0	17:45	?	00:00:00	[kworker/R-kt

```

root      2030      1 0 17:45 ?      00:00:00 /usr/lib/syst
ec2-user  2035      1 0 17:45 ?      00:00:00 /usr/lib/syst
ec2-user  2037     2035 0 17:45 ?      00:00:00 (sd-pam)
ec2-user  2044     1962 0 17:45 ?      00:00:05 sshd-session:
ec2-user  2045     2044 0 17:45 pts/0      00:00:00 -bash
root      28029     2 0 18:45 ?      00:00:00 [kworker/0:1-
root      28351     2 0 18:46 ?      00:00:00 [kworker/u8:3
root      28373     2 0 18:46 ?      00:00:00 [kworker/1:0-
root      29207     2 0 18:57 ?      00:00:00 [kworker/u8:1
root      29694     2 0 19:02 ?      00:00:00 [kworker/0:2-
root      29764     2 0 19:03 ?      00:00:00 [kworker/1:2-
root      29837     1 0 19:04 ?      00:00:00 /usr/sbin/htt
apache    29838     29837 0 19:04 ?      00:00:00 /usr/sbin/htt
apache    29839     29837 0 19:04 ?      00:00:00 /usr/sbin/htt
apache    29841     29837 0 19:04 ?      00:00:00 /usr/sbin/htt
apache    29842     29837 0 19:04 ?      00:00:00 /usr/sbin/htt
apache    30075     29837 0 19:06 ?      00:00:00 /usr/sbin/htt
root      30245     2030 0 19:08 ?      00:00:00 systemd-userw
root      30246     2030 0 19:08 ?      00:00:00 systemd-userw
root      30249     2030 0 19:08 ?      00:00:00 systemd-userw
root      30307     2 0 19:09 ?      00:00:00 [kworker/1:1]
ec2-user  30308     2045 0 19:09 pts/0      00:00:00 ps -ef

```

```
[ec2-user@ip-172-31-34-234 ~]$ top
```

```

top - 19:10:07 up 1:25, 1 user, load average: 0.00, 0.00, 0.0
Tasks: 110 total, 1 running, 109 sleeping, 0 stopped, 0 zo
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi,
MiB Mem : 913.2 total, 308.3 free, 202.3 used, 402.5
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 572.8

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	root	20	0	173692	18024	10984	S	0.0	1.9
2	root	20	0	0	0	0	S	0.0	0.0
3	root	20	0	0	0	0	S	0.0	0.0
4	root	0	-20	0	0	0	I	0.0	0.0
5	root	0	-20	0	0	0	I	0.0	0.0
6	root	0	-20	0	0	0	I	0.0	0.0
7	root	0	-20	0	0	0	I	0.0	0.0
8	root	0	-20	0	0	0	I	0.0	0.0
10	root	0	-20	0	0	0	I	0.0	0.0
12	root	20	0	0	0	0	I	0.0	0.0
13	root	0	-20	0	0	0	I	0.0	0.0
14	root	20	0	0	0	0	S	0.0	0.0
15	root	20	0	0	0	0	I	0.0	0.0
16	root	20	0	0	0	0	S	0.0	0.0
17	root	20	0	0	0	0	S	0.0	0.0
18	root	rt	0	0	0	0	S	0.0	0.0
19	root	20	0	0	0	0	S	0.0	0.0
20	root	20	0	0	0	0	S	0.0	0.0
21	root	rt	0	0	0	0	S	0.0	0.0
22	root	20	0	0	0	0	S	0.0	0.0
24	root	0	-20	0	0	0	I	0.0	0.0

```

27 root      20    0      0      0      0 S    0.0  0.0
28 root      0 -20      0      0      0 I    0.0  0.0
29 root      20    0      0      0      0 I    0.0  0.0
30 root      20    0      0      0      0 I    0.0  0.0
31 root      20    0      0      0      0 I    0.0  0.0
32 root      20    0      0      0      0 S    0.0  0.0
34 root      20    0      0      0      0 S    0.0  0.0
35 root      20    0      0      0      0 S    0.0  0.0
37 root      0 -20      0      0      0 I    0.0  0.0
38 root      20    0      0      0      0 S    0.0  0.0
39 root      39   19      0      0      0 S    0.0  0.0
[ec2-user@ip-172-31-34-234 ~]$ pgrep httpd
29837
29838
29839
29841
29842
30075
[ec2-user@ip-172-31-34-234 ~]$ ps -p 29838 -f
UID          PID    PPID  C STIME TTY          TIME CMD
apache      29838    29837  0 19:04 ?           00:00:00 /usr/sbin/htt
[ec2-user@ip-172-31-34-234 ~]$ sudo systemctl stop httpd
[ec2-user@ip-172-31-34-234 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enab>
   Active: inactive (dead) since Tue 2026-08-25 19:11:51 UTC; >
   Duration: 6min 57.501s
   Docs: man:httpd.service(8)
   Process: 29837 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROU>
   Main PID: 29837 (code=exited, status=0/SUCCESS)
   Status: "Total requests: 4; Idle/Busy workers 100/0;Request>
   CPU: 508ms

Aug 25 19:04:53 ip-172-31-34-234.ec2.internal systemd[1]: Starti>
Aug 25 19:04:53 ip-172-31-34-234.ec2.internal systemd[1]: Starte>
Aug 25 19:04:53 ip-172-31-34-234.ec2.internal httpd[29837]: Serv>
Aug 25 19:11:50 ip-172-31-34-234.ec2.internal systemd[1]: Stoppi>
Aug 25 19:11:51 ip-172-31-34-234.ec2.internal systemd[1]: httpd.>
Aug 25 19:11:51 ip-172-31-34-234.ec2.internal systemd[1]: Stoppe>

[ec2-user@ip-172-31-34-234 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-34-234 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enab>
   Active: active (running) since Tue 2026-08-25 19:12:30 UTC; >
   Docs: man:httpd.service(8)
   Main PID: 30448 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Request>
   Tasks: 177 (limit: 1059)
   Memory: 13.7M
   CPU: 83ms

```

```
[ec2-user@ip-172-31-34-234 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-34-234 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-08-25 19:12:30 UTC; 1min 1s ago
     Docs: man:httpd.service(8)
   Main PID: 30448 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Request queue: 0"
     Tasks: 177 (limit: 1059)
   Memory: 13.7M
     CPU: 83ms
   CGroup: /system.slice/httpd.service
           └─30448 /usr/sbin/httpd -DFOREGROUND
             └─30449 /usr/sbin/httpd -DFOREGROUND
               └─30450 /usr/sbin/httpd -DFOREGROUND
                 └─30453 /usr/sbin/httpd -DFOREGROUND
                   └─30465 /usr/sbin/httpd -DFOREGROUND

Aug 25 19:12:30 ip-172-31-34-234.ec2.internal systemd[1]: Starting Apache HTTP Server: httpd.service.
Aug 25 19:12:30 ip-172-31-34-234.ec2.internal httpd[30448]: Server started.
Aug 25 19:12:30 ip-172-31-34-234.ec2.internal systemd[1]: Started Apache HTTP Server: httpd.service.

[ec2-user@ip-172-31-34-234 ~]$
```

Search and Text Processing

Commands used:

```
touch students.txt
nano students.txt
grep "vishal" students.txt
grep -i "a" students.txt
grep -i -c "a" students.txt
sort students.txt
sort -k2 students.txt
```

Screenshot:

```
[ec2-user@ip-172-31-34-234 ~]$ echo 'Q11'
Q11
[ec2-user@ip-172-31-34-234 ~]$ cd linuxexam
[ec2-user@ip-172-31-34-234 linuxexam]$ touch students.txt
[ec2-user@ip-172-31-34-234 linuxexam]$ nano students.txt
[ec2-user@ip-172-31-34-234 linuxexam]$ grep "vishal" students.txt
2. vishal age:21
[ec2-user@ip-172-31-34-234 linuxexam]$ grep -i "a" students.txt
1. deepak age:22
2. vishal age:21
3. gaju age: 23
4. shubham age:25
5. nitesh age:22
6. vikas age:20
7.nilesh age:22
8.mane age:30
9. om age:22
10. sandeep age:23
[ec2-user@ip-172-31-34-234 linuxexam]$ grep -i -c "a" students.txt
10
[ec2-user@ip-172-31-34-234 linuxexam]$ sort students.txt

1. deepak age:22
10. sandeep age:23
2. vishal age:21
3. gaju age: 23
4. shubham age:25
5. nitesh age:22
6. vikas age:20
7.nilesh age:22
8.mane age:30
9. om age:22
[ec2-user@ip-172-31-34-234 linuxexam]$ sort -k2 students.txt

7.nilesh age:22
8.mane age:30
1. deepak age:22
3. gaju age: 23
5. nitesh age:22
9. om age:22
10. sandeep age:23
4. shubham age:25
6. vikas age:20
2. vishal age:21
[ec2-user@ip-172-31-34-234 linuxexam]$ grep -i "a" students.txt
1. deepak age:22
2. vishal age:21
3. gaju age: 23
4. shubham age:25
5. nitesh age:22
```

```
5. nitesh age:22
6. vikas age:20
7.nilesh age:22
8.mane age:30
9. om age:22
10. sandeep age:23
[ec2-user@ip-172-31-34-234 linuxexam]$ grep -i -E '^[0-9]+ [^]*a[^]* ' students.txt
```

Q12. Linux Networking

Commands used:

```
hostname
ip addr
ip link
ip route
ping google.com
ping -c 4 8.8.8.8
curl http://<server-ip>
nslookup <domain>
sudo ss -tulnp
```

Screenshot:

```
[ec2-user@ip-172-31-34-234 linuxexam]$ echo 'Q12'
Q12
[ec2-user@ip-172-31-34-234 linuxexam]$ hostname
ip-172-31-34-234.ec2.internal
[ec2-user@ip-172-31-34-234 linuxexam]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0e:e3:6b:ba:da:a9 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-0b871e44219ef8b70
    altname device-number-0.0
    inet 172.31.34.234/20 metric 512 brd 172.31.47.255 scope global dynamic ens5
        valid_lft 2577sec preferred_lft 2577sec
    inet6 fe80::ce3:6bff:feba:daa9/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-34-234 linuxexam]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 0e:e3:6b:ba:da:a9 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-0b871e44219ef8b70
    altname device-number-0.0
[ec2-user@ip-172-31-34-234 linuxexam]$ cd
[ec2-user@ip-172-31-34-234 ~]$ ip route
default via 172.31.32.1 dev ens5 proto dhcp src 172.31.34.234 metric 512
172.31.0.2 via 172.31.32.1 dev ens5 proto dhcp src 172.31.34.234 metric 512
172.31.32.0/20 dev ens5 proto kernel scope link src 172.31.34.234 metric 512
172.31.32.1 dev ens5 proto dhcp scope link src 172.31.34.234 metric 512
[ec2-user@ip-172-31-34-234 ~]$ ping google.com
PING google.com (192.178.155.138) 56(84) bytes of data.
64 bytes from yuiadrs-in-f138.1e100.net (192.178.155.138): icmp_seq=1 ttl=106 time=2.59 ms
64 bytes from yuiadrs-in-f138.1e100.net (192.178.155.138): icmp_seq=2 ttl=106 time=2.62 ms
64 bytes from yuiadrs-in-f138.1e100.net (192.178.155.138): icmp_seq=3 ttl=106 time=2.64 ms
64 bytes from yuiadrs-in-f138.1e100.net (192.178.155.138): icmp_seq=4 ttl=106 time=2.62 ms
64 bytes from yuiadrs-in-f138.1e100.net (192.178.155.138): icmp_seq=5 ttl=106 time=2.63 ms
```

```

2.61 ms
64 bytes from yuiadrs-in-f138.1e100.net (192.178.155.138): icmp_seq=8 ttl=106 time=
2.78 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7009ms
rtt min/avg/max/mdev = 2.593/2.642/2.776/0.052 ms
[ec2-user@ip-172-31-34-234 ~]$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=1.59 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=1.63 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=1.56 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=117 time=1.64 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.556/1.604/1.641/0.033 ms
[ec2-user@ip-172-31-34-234 ~]$ curl http://3.95.0.98
student name: hanumant umatawade
roll number: 30
course name: mca
"linux practical examination"

[ec2-user@ip-172-31-34-234 ~]$ nslookup playerspeak.in
Server:      172.31.0.2
Address:     172.31.0.2#53

Non-authoritative answer:
Name:   playerspeak.in
Address: 54.237.76.2

[ec2-user@ip-172-31-34-234 ~]$ sudo ss -tulnp
Netid State  Recv-Q Send-Q           Local Address:Port  Peer Address:Port
Process
udp    UNCONN  0      0                127.0.0.1:323      0.0.0.0:*
users:(("chronyd",pid=1918,fd=5))
udp    UNCONN  0      0      172.31.34.234%ens5:68  0.0.0.0:*
users:(("systemd-network",pid=1713,fd=17))
udp    UNCONN  0      0                [::1]:323         [::]:*
users:(("chronyd",pid=1918,fd=6))
udp    UNCONN  0      0      [fe80::ce3:6bff:feba:daa9]%ens5:546 [::]:*
users:(("systemd-network",pid=1713,fd=19))
tcp    LISTEN  0      128                0.0.0.0:22        0.0.0.0:*
users:(("sshd",pid=1889,fd=7))
tcp    LISTEN  0      128                [::]:22          [::]:*
users:(("sshd",pid=1889,fd=8))
tcp    LISTEN  0      511                  *:80              *:
users:(("httpd",pid=30465,fd=4),("httpd",pid=30453,fd=4),("httpd",pid=30450,fd=4),
("httpd",pid=30448,fd=4))
[ec2-user@ip-172-31-34-234 ~]$ |

```